

Data Mining Project - Problem Statement Outline

Residential Electricity Consumption Pattern Mining

1. Problem Statement

Can data-mining techniques discover distinct residential electricity-consumption patterns and classify a new consumer into the appropriate consumption group?

Residential consumers can have very different electricity-use behaviors. The project aims to discover these hidden patterns from historical smart-meter data instead of defining the groups manually.

2. Main Objectives

- Obtain and understand a suitable Indian residential smart-meter dataset.
- Clean and preprocess the electricity-consumption data.
- Transform raw readings into meaningful household consumption profiles.
- Use clustering to discover groups of consumers with similar consumption patterns.
- Interpret the characteristics of the discovered groups.
- Use classification models to assign new consumption profiles to the discovered groups.
- Evaluate and compare the performance of the selected models.

3. Proposed Workflow

Stage	What we do
1. Data	Collect Indian household smart-meter readings with meter/household IDs and timestamps.
2. Preprocessing	Handle missing values, duplicates, inconsistent records and data types.
3. Feature construction	Convert time-based readings into comparable household consumption profiles.
4. Clustering	Apply K-Means as the primary method; compare with Hierarchical Clustering and/or GMM if useful.
5. Interpretation	Study the consumption characteristics of each discovered cluster.
6. Classification	Use KNN, Decision Tree and/or SVM to classify new consumption profiles.
7. Evaluation	Use clustering metrics such as Silhouette/Davies-Bouldin and classification metrics such as accuracy, precision, recall and F1-score.

4. How the Data Will Be Represented

Raw data: Meter/household ID + date/time + electricity usage (and other available measurements).

Transformed data: Each household becomes a profile containing consumption values across consistent time periods, such as hourly or daily intervals.

Example: Household 101 → [00:00 usage, 01:00 usage, ..., 23:00 usage]. K-Means can then group households with similar consumption shapes.

5. Data Mining Concepts Used

Concept	Role in the project
K-Means	Primary clustering method for discovering consumer groups.
Hierarchical Clustering	Optional comparison to examine whether similar groups are obtained.
GMM / EM	Optional probabilistic clustering comparison if the dataset supports it.
KNN	Classify a new profile using similarity to previously classified profiles.
Decision Tree	Interpretable classification and identification of influential features.
SVM	Classification comparison against other supervised models.
Preprocessing & Evaluation	Cleaning, scaling/normalization, feature engineering, validation and performance measurement.

6. Expected Outcome

The system should discover meaningful residential electricity-consumption profiles and classify an unseen household/profile into one of those groups. The final group labels will come from the data, rather than being assumed in advance.

7. Possible Innovation

A manageable extension is to compare multiple clustering methods and then test whether the discovered groups can be reliably recognized by supervised classifiers. Another option is to interpret which time periods or consumption features distinguish the groups most strongly.

8. Research Basis

Primary reference: S. Palaniappan, S. Karuppanan, D. Velusamy, "Categorization of Indian residential consumers electrical energy consumption pattern using clustering and classification techniques," *Energy*, 289, 129992, 2024. DOI: 10.1016/j.energy.2023.129992.

The research problem is to categorize Indian residential consumers according to their electricity-consumption patterns using clustering and classification techniques.

9. Dataset Direction

For implementation, use a publicly accessible Indian smart-meter dataset with multiple households/meters, timestamps and electricity-consumption measurements. The exact dataset should be verified before the final project commitment so that it supports household-level profile construction.

10. Day-1 Target

- Confirm the dataset and inspect its columns and units.
- Check missing values, duplicates and the number of households/meters.
- Create a first household-level consumption profile.
- Run a basic K-Means experiment and visualize the initial clusters.

Important: The project is based on the research problem, but the implementation dataset does not have to be the private dataset used in the 2024 paper.